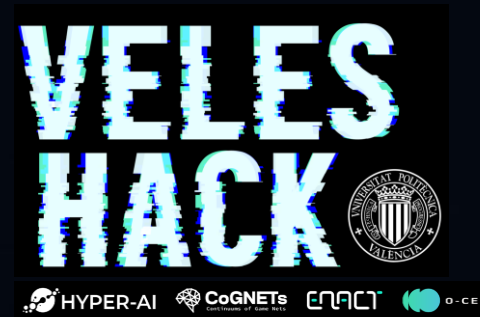


GreenCharge

Autonomous, Carbon-Aware EV Charging & Dynamic ENACT Adaptation

ENACT VelesHack 2026 | Sovereign Dataspaces & Adaptive Cloud-Native Execution



Funded by
the European Union

GitHub repo

- **GitHub Repository:** <https://github.com/Adelin12378/ENACT-VELESHACK-2026>
- **Project Stack:** Java 21, Spring Boot 3 microservice, Helm packaging (chart/), and containerized automated test suite.
- **ENACT Infrastructure:** Local 3-node Kind Kubernetes cluster (enact-dev) preconfigured with Cilium eBPF CNI, Hubble UI, and Kepler energy exporter.
- **Exposed APIs:** EV route scoring (/route), fleet status (/chargers), sovereign carbon feed ingest (/dataspace/carbon), and dynamic adaptation (/adaptation).

Summary

- **The Problem:** Traditional EV routing ignores dynamic municipal grid carbon intensity (55–240 gCO₂/kWh across districts), resulting in unnecessary carbon emissions and peak fossil grid stress.
- **The GreenCharge Solution:** Dynamically routes EV drivers to the greenest available municipal charger in real time, updating recommendations as grid carbon intensities swing.
- **Sovereign Dataspace Integration:** Ingests live utility grid-carbon-intensity telemetry under a sovereign dataspace contract; transitions UI from mock to live feed with zero app code changes.
- **Policy-Governed Deployment:** Packaged via Helm and deployed on green-powered compute nodes governed by declarative ENACT runtime policies.

Highlights

- **Application Policy Model (APM):** Declarative `policymodel.yml` enforces placement and sustainability rules: Region eu-west (Hard), Availability ≥ 0.90 (Hard), and Green Energy Mix ≥ 0.60 (Soft).
- **Application Controller (AC) Adaptation:** Engineered `AdaptationService` evaluating host `ResourceMetrics`; dynamically recommends automated elasticity actions (`scale_up` / `scale_down` / `no_action`).
- **Green Scheduling Enforcement:** Workload scheduled onto green-compliant worker nodes (`enact-dev-worker`) via ENACT APPLPM controller manager and custom `RuntimePolicy` CRD.
- **Full Observability & Validation:** Monitored via TDCME Monitor API, Prometheus, and Grafana; node power tracked with Kepler and network flows visualized via Cilium Hubble.